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Kitchen waste bone-driven enzyme-induced calcium phosphate precipitation under microgravity for space biocementation

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ABSTRACT

This study validates the feasibility of extracting calcium and phosphorus from kitchen waste bones for crude enzyme-induced calcium phosphate precipitation (EICPP) under both normal and microgravity conditions. The experimental results demonstrate no significant differences in the degree of reaction and characteristics of precipitation between these environments. By leveraging local resources, reducing material transport costs, and addressing waste management challenges, this research underscores the potential for extraterrestrial construction, thereby enhancing sustainability in space environments. These findings offer promising insights for the application of space biocementation, particularly during the expansion phase of human settlements.

1. Introduction

With the advancement of human exploration into outer space, the objectives have gradually shifted from initial exploration and experimentation to the establishment of enduring bases or colonies on the Moon, Mars, and other celestial bodies. However, achieving this grand goal presents numerous challenges, one of which is the prohibitively high cost of transporting materials for construction and maintenance (Isachenkov et al., 2021; Parkinson, 1991; Soureshjani et al., 2023; Toklu & Akpinar, 2022; Wang et al., 2022). This makes the task of transporting resources from Earth to space a formidable challenge. Additionally, frequent rocket launches impose an extra burden on the environment (Dallas et al., 2020; Koroleva, 2018; Maloney et al., 2022; Ross & Jones, 2022; Ryan et al., 2022). Consequently, finding alternative solutions to reduce the need for material transport from Earth has become an urgent issue for scientists. In response to this challenge, In-Situ Resource Utilization (ISRU) technology has emerged and gained increasing attention. The core concept of ISRU technology is to utilize local resources on the target celestial body to reduce dependence on Earth's resources (Anand et al., 2012; Hecht et al., 2021; Hoffman, 2022; Palos et al., 2020; Muscatello, 2020). This encompasses not only the production of construction materials from local resources but also the acquisition of essential resources such as water, oxygen, and fuel (Kruyer et al., 2021; Lomax, 2022; Ma, 2022; Ross et al., 2023; Snehal et al., 2024). By effectively utilizing in-situ resources, the self-sufficiency of extraterrestrial bases can be enhanced, significantly lowering the economic costs of space construction.

As part of ISRU technology, biocementation is being extensively studied (Dikshit et al., 2021; Gleaton et al., 2019, 2022; Santomartino et al., 2023; Zuo et al., 2023). This technique utilizes microorganisms or enzymes to induce mineral precipitation in a substrate, thereby enhancing the geotechnical properties of the soil. Biocementation has already demonstrated its broad application potential in terrestrial infrastructure projects (Bian et al., 2024; Cuccurullo et al., 2022; Dong & Liu, 2022; Singh et al., 2021; Xu et al., 2024). Among various biocementation methods, the use of specialized acidic ureases or acid-tolerant urease-producing bacteria can induce the formation of calcium phosphate biocement in neutral or slightly acidic pH systems. This approach helps convert gaseous ammonia into its ionic form, thereby reducing ammonia emissions (Avramenko et al., 2023; Gowthaman et al., 2021; Ivanov et al., 2019). However, the need for acid-resistant bacteria or enzymes somewhat limits its development. This study explores an innovative approach by using inexpensive and readily available soybean crude enzymes (He et al., 2022; Liu et al., 2024; Xue et al., 2024; Cui et al., 2024), to utilize kitchen waste bones as sources of calcium and phosphorus in the enzyme-induced calcium phosphate precipitation (EICPP) technique, with the aim of evaluating its potential application in microgravity environments.

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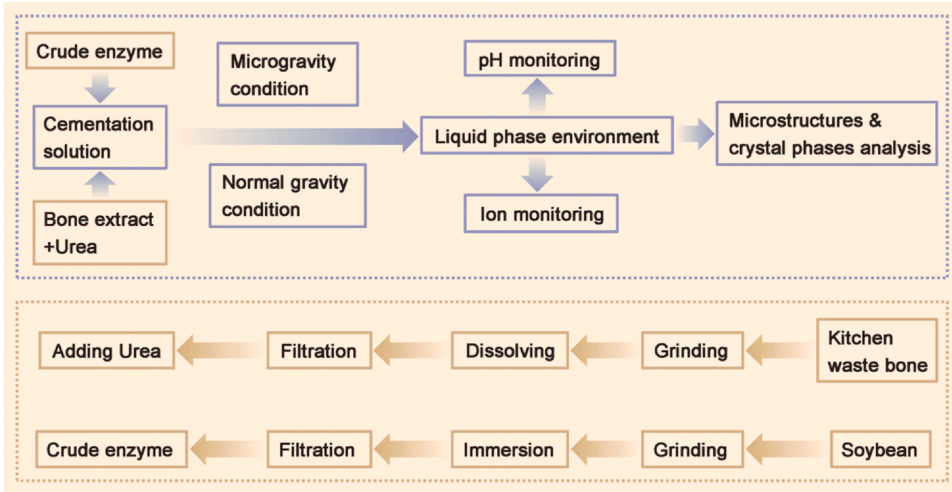

Fig. 1. Flow chart of the experiment.

In this study, as shown in the Fig. 1, calcium and phosphorus were extracted from kitchen waste bones using acid. In the presence of soybean crude enzymes, which hydrolyze urea and induce a rise in pH, calcium and phosphorus in the cementation solution effectively precipitated as calcium phosphate. The experimental results indicate that under both terrestrial and microgravity conditions, the degree of EICPP reaction, the uniformity of calcium phosphate precipitation, and the microstructural characteristics of the precipitates did not exhibit significant differences. These results provide support for the application of EICPP technology in extraterrestrial construction. This technology holds significant promise for the expansion phase of extraterrestrial settlements. As extraterrestrial colonies expand and populations increase, the generation of organic waste such as animal bones will rise substantially. Utilizing these waste materials as sources of calcium and phosphorus in EICPP can not only help reduce the cost of construction materials in space but also effectively address waste management challenges, thereby enhancing the sustainability of resource utilization in space environments.

2. Materials and methods

2.1. Waste bone and crude enzyme and extraction

In this study, kitchen waste pig bones, as shown in Fig. 2A, were utilized as a sustainable source for the extraction of the curing solution. X-ray fluorescence (XRF) analysis obtained using the Hitachi SEA1200VX X-ray fluorescence analyzer, depicted in Fig. 2E, revealed that the bones are rich in calcium and phosphorus, while X-ray diffraction (XRD) analysis, shown in Fig. 2F, further identified the primary component as hydroxyapatite. The process of calcium and phosphorus extraction from the bones involved several steps. First, the cleaned bones were dried at room temperature and then ground into particles smaller than 150 μm, as shown in Fig. 2B, to prepare them for the subsequent steps. A 40 g sample of bone powder was then dissolved in 100 mL of 2 mol/L HCl solution, as illustrated in Fig. 2C, with stirring at 1500 r/min. The dissolution process was monitored through pH and calcium ion concentration, aiming to ensure its progression, as indicated in Fig. 2G, before the solution was filtered, as shown in Fig. 2D. The primary hydroxyapatite in the bones was acid hydrolyzed according to Eq. (1). The solution was diluted and adjusted to reach a pH of (3.6 ± 0.2) and a calcium ion concentration of (8500 ± 200) ppm. Finally, urea was added to the bone extract solution according to the different groups listed in Table 1.

$$Ca_{10}(PO_4)_6(OH)_2 + 8\,HCl \rightarrow 10\,Ca^{2+} + 6\,HPO_4{}^{2-} + 8\,Cl^- + 2\,H_2O \quad (1)$$

The extraction of crude soybean (*Glycine max*) urease in this experiment followed a simple procedure: commercially available dry soybeans were ground and passed through a 150 μm sieve. The soybean powder was added to distilled water at a concentration of 50 g/L, stirred for 30 min, and then centrifuged to collect the supernatant. The crude soybean urease solution, which required no further purification and had an activity of approximately 11.2 U, was obtained through this cost-effective and practical method of extraction from soybeans. In this study, the cementation solution was prepared by mixing equal volumes of the crude enzyme solution and the bone extract. The crude enzyme and bone extract were only combined immediately before used to prevent premature reactions.

2.2. Liquid phase experiment

To monitor the reaction progress in the liquid-phase environment and to understand the effects of microgravity on crude soybean enzyme-induced calcium phosphate precipitation, experiments were conducted under both microgravity and normal gravity conditions, as shown in Table 1. Prior to the experiments, the crude enzyme and bone extract solutions, prepared as described above, were stored in a refrigerator at 4 °C to pre-cool the solutions. The reason for cooling process is that low temperature can delay the start of the experiment by inhibiting the activity of the enzyme, and as much as possible, the reaction can be temporarily delayed during the operation steps such as the injection of the cementation solution before the microgravity test machine provides microgravity to the sample, thereby ensuring that the experimental process can be carried out in a microgravity environment to the greatest extent possible.

Once cooled, the solutions were quickly mixed and transferred into the appropriate containers. For the samples subjected to microgravity, they were placed in the Kitagawa Zeromo CL-1000 microgravity clinostat machine, which simulates a microgravity environment at 10^{-3} times Earth's gravity by rotating the samples in three-dimensional space (Brungs et al., 2016; Kim et al., 2017, 2023; Oluwafemi & Neduncheran, 2022; Wuest, 2014). The microgravity simulator was housed in a temperature-controlled chamber set to 25 °C. The samples under normal gravity conditions followed the same protocol to ensure that all experimental variables, aside from gravity, were consistent. After 24 h of reaction, the samples were collected and analyzed. In this study, after mixing equal volumes of the bone extract solution and crude enzyme, the initial calcium ion concentration was 4250 ppm. Calcium ion concentration was monitored using the Horiba LAQUAtwin Ca-11 compact calcium ion meter, and pH was measured using the Horiba LAQUA F-71 benchtop pH meter. Additionally, ammonia/

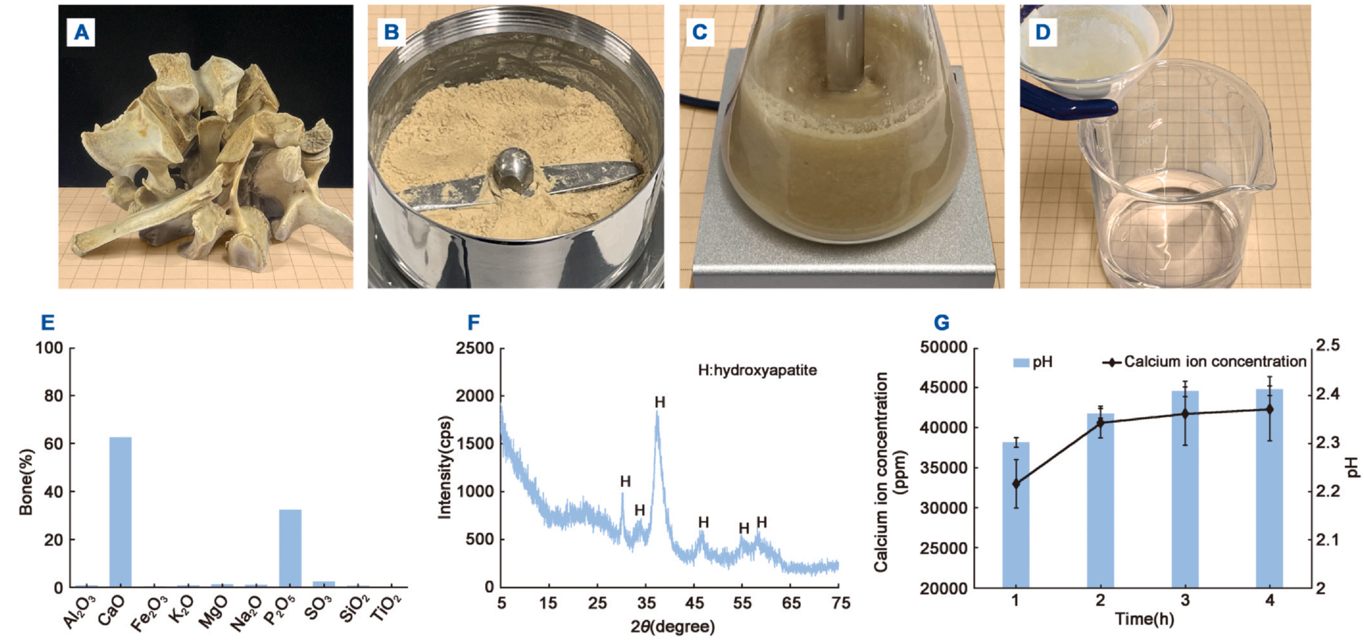

Fig. 2. **(A)** Kitchen waste bone used; **(B–D)** Processes of grinding, dissolving, and filtering during the preparation of bone extracts; **(E)** XRF analysis of the bone powder; **(F)** Corresponds to XRD; **(G)** pH and calcium ions during bone dissolution.

Table 1
Formulations of the samples.

Case name	Gravity condition	Urea/Ca^{2+} ratio
MA		0
MB		0.25
MC	Microgravity	0.5
MD		0.75
ME		1
NA		0
NB		0.25
NC	Normal gravity	0.5
ND		0.75
NE		1

ammonium concentration was determined using the indophenol spectrophotometric method based on the Berthelot reaction (Cogan et al., 2014; Jain et al., 2021; Li et al., 2020; Remiszewska, 2019; Molins-Legua et al., 2006).

2.3. Precipitation characterization

After 24 h, the samples were centrifuged to remove the liquid, and the precipitates were rinsed with distilled water, followed by drying in an oven at 60 °C for 48 h in preparation for further analysis. The morphology of the precipitates was examined via scanning electron microscopy (SEM) using a JEOL JSM-IT200, and energy-dispersive spectroscopy (EDS) was conducted for elemental analysis. Additionally, the crystalline structure of the precipitates and the bone was characterized using the Rigaku MiniFlex benchtop X-ray diffractometer equipped with a Co Kα radiation source, with a scanning rate of 6.5°/min.

3. Results and discussion

3.1. Reaction monitoring and analysis

Fig. 3A–B show the pH values in the liquid-phase environment after 24 h of reaction. It is evident that pH increases with the rising of urea content in the samples. Since the amount of added enzyme remains constant, an increase in urea implies a greater degree of hydrolysis, as described by Eq. (2) and Eq. (3), leading to more hydrolysis products and consequently a larger rise in pH. Notably, even in sample ME and NE, which exhibited the highest reaction extent, the pH remained close to neutral. This near-neutral pH facilitates the conversion of ammonia into its ionic form, favoring ammonia control and ammonium recovery. This phenomenon can be attributed to the residual acid from bone dissolution, which contributes to the relatively low initial pH, as well as the buffering effect of the significant protein content in the crude enzyme (Krishnan et al., 2023).

$$CO(NH_2)_2 + 2\,H_2O \xrightarrow{\text{urease}} H_2CO_3 + 2\,NH_3 \tag{2}$$

$$NH_3 + H_2O \leftrightarrow OH^- + NH_4^+ \tag{3}$$

Fig. 3C–D illustrate the calcium ion concentration in the cementation solution after 24 h of reaction. As shown, the calcium ion concentration decreases as the urea content increases. The rise in pH resulting from increased urea content reduces calcium solubility, which, in turn, promotes precipitation as outlined in the corresponding equations, consuming calcium ions. It is noteworthy that even in the sample MA and NA, where no urea was added, the pH slightly increased compared to the bone extract solution due to the addition of the crude enzyme, leading to a certain degree of precipitation and calcium ion consumption.

Fig. 3E–F show the concentration of ammonia/ammonium in the consolidation solution after 24 h. Since ammonia/ammonium is a product of urea hydrolysis, its concentration increases proportionally with the urea content. Throughout the experiments, no significant differences were observed between the microgravity and normal gravity conditions in terms of pH, calcium ion concentration, or ammonia levels, suggesting the potential applicability of this method in future space applications.

3.2. Precipitation analysis

As illustrated in Fig.3(G–J), the SEM images further confirm that the morphology of precipitates in both microgravity and normal gravity environments showed no significant differences, with the precipitates

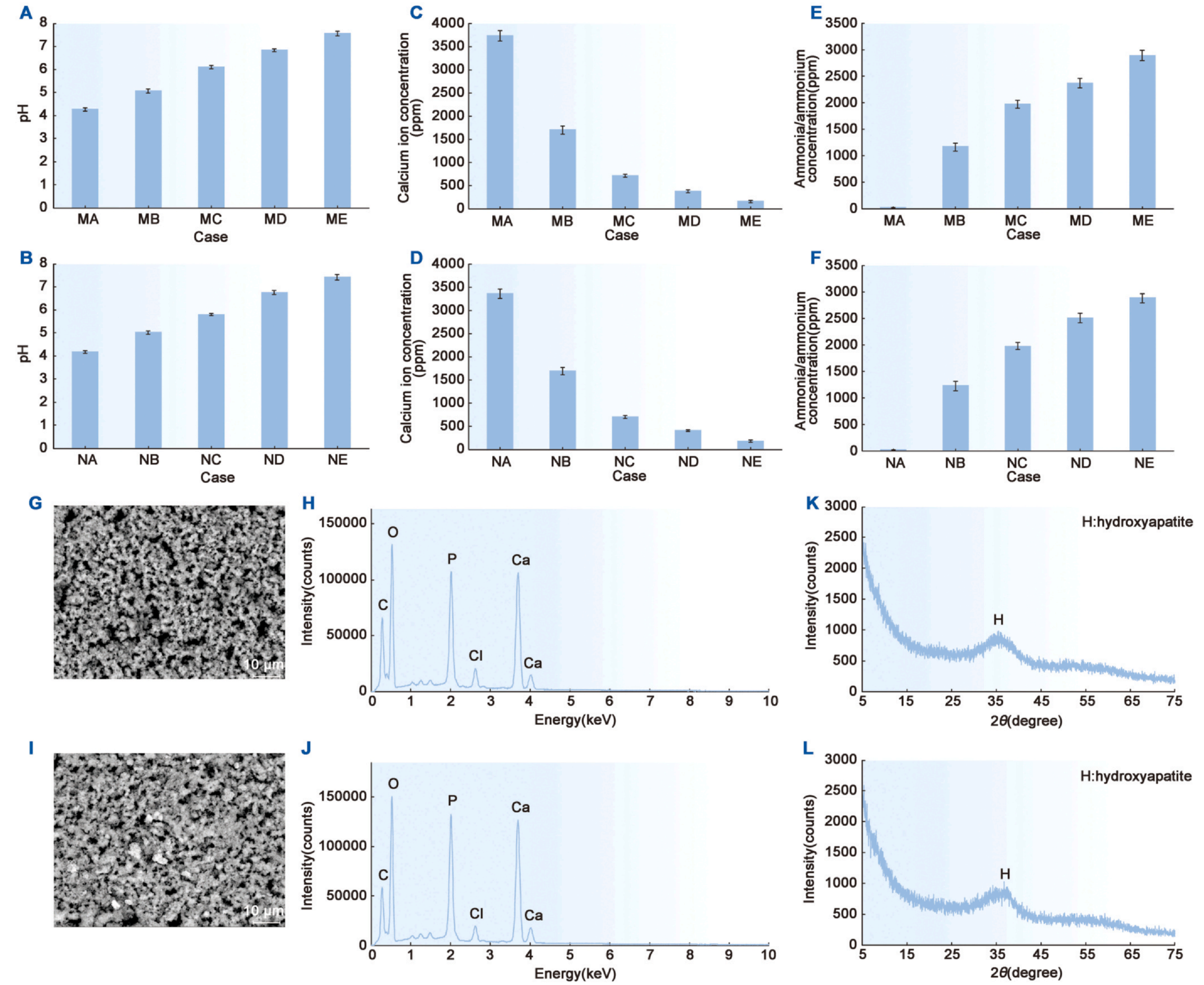

Fig. 3. **(A)** and **(B)** show the pH monitoring of the liquid phase experiment under microgravity and normal gravity conditions, respectively; **(C–D)** Calcium ion concentration monitoring of the liquid phase experiment under microgravity and normal gravity conditions, respectively; **(E–F)** Ammonia/ammonium concentration monitoring of the liquid phase experiment under microgravity and normal gravity conditions, respectively; **(G–H)** SEM and EDS analysis of the precipitate under microgravity conditions; **(I–J)** SEM and EDS analysis of the precipitate under normal gravity conditions; **(K–L)** XRD analysis of the precipitate under microgravity and normal gravity conditions, respectively.

being generally small in size. When precipitate particles are small, gravity-driven convection is minimal, which explains why the reaction rates under both gravity conditions were quite similar.

$$10\ Ca^{2+} + 6\ PO_4^{3-} + 2\ OH^- \rightarrow Ca_{10}(PO_4)_6(OH)_2 \quad (4)$$

The EDS and XRD analyses also revealed no significant variations. As described in Eq. (4), the EDS results in Fig. 3(G–J) indicated that the precipitates primarily consisted of calcium, phosphorus, oxygen, and carbon. The presence of chlorine in the EDS data stems from the hydrochloric acid used during bone dissolution, and despite thorough washing, a small amount of chloride ions remained in the precipitates. XRD analysis in Fig. 3(K–L) identified hydroxyapatite ($Ca_{10}(PO_4)_6(OH)_2$) as the dominant crystalline phase, corroborating the EDS results. Previous studies have demonstrated that hydroxyapatite, as the thermodynamically most stable crystalline phase (Arokiasamy et al., 2022; Borum-Nicholas & Wilson, 2003; Jin et al., 2015; Koutsopoulos, 2001; Pan & Darvell, 2010), possesses higher mechanical strength than brushite (Charrière et al., 2001). The transformation of brushite into hydroxyapatite typically requires high pH and elevated temperatures (Avramenko et al., 2024; da Rocha et al., 2018; Furutaka et al., 2006; Štulajterová & Medvecký, 2008; Zhang et al., 2019). In this study, however, the precipitated crystalline phase was directly hydroxyapatite, which may be partly due to the high concentration of soybean proteins in the crude enzyme extract, promoting hydroxyapatite formation, and partly due to the drying process in a 60 °C oven, which likely favored its crystallization. Further investigation is needed to explore these factors, along with solidification experiments incorporating regolith or its simulants. Additionally, the distribution of precipitates has a significant impact on the consolidation effect (Almajed et al., 2020; DeJong et al., 2010; Liu et al., 2021; Ma, 2021; Xiao, 2019). Under microgravity conditions, the precipitation patterns and distribution of biocement may differ substantially from those on Earth. Therefore, further assessment, optimization, and implementation of the desired biocement consolidation effect in microgravity or low-gravity environments is crucial. At the same time, grouting methods in space environments require special design to ensure the effective delivery of reaction substrates and enzymes.

3.3. Developmental outlook and potential applications analysis

As humanity ventures deeper into outer space and gradually begins establishing permanent bases and colonies on the Moon and Mars, the significance of certain technologies becomes more pronounced, especially during the expansion phase following long-term extraterrestrial settlement. With the gradual growth of extraterrestrial colonies and the increase in both human and animal populations, there will inevitably be a significant generation of waste, including animal excrement and bone waste. Effective waste management is critical to ensuring resource sustainability in space missions. Integrating EICPP technology into long-term extraterrestrial settlement plans allows for the utilization of such waste as sources of urea, calcium, and phosphorus in construction processes. This not only helps reduce the costs of interstellar construction but also addresses the challenge of waste disposal in space bases, thereby enhancing the sustainability of resource utilization. Given the challenges of water extraction on the Moon and Mars, despite the evidence of the existence of water resources, it is still crucial to minimize water consumption during the production process or develop water recycling technologies, such as combining the recycling of urine. The acid used to dissolve the discarded bones is more likely to be sulfuric acid in the future, considering the ISRU-based principle, because both the Moon and Mars are rich in sulfur. Mars, in particular, is considered a sulfur-rich planet where the sulfur cycle is an important geochemical process (Ding et al., 2014; Gaillard et al., 2013; Halevy et al., 2007; King & McLennan, 2010; Righter et al., 2009). The neutral pH of the EICPP process helps minimize ammonia emissions and promotes the recycling of ammonium ions. It also enables the use of recycled ammonium ions for plant growth, further advancing resource circularity (Yan et al., 2024). This capability is of great importance to life support and resource management systems in extraterrestrial environments.

One promising application of EICPP in space base construction is the utilization of lunar and Martian regolith for building and other extraterrestrial geotechnical applications as shown in Fig. 4. The high levels of radiation on the Moon and Mars make EICPP technology, combined with local regolith, a potential solution for creating radiation barriers that offer sustainable protection for interplanetary bases (Montes, 2015). Additionally, EICPP can stabilize the regolith on the Moon and Mars, reducing dust accumulation and erosion. The surfaces of these celestial bodies are covered with fine dust and particles, which not only pose challenges for construction but also threaten equipment and human health (Belobrajdic et al., 2021; Borisova, 2019; Cain, 2010; Khan-Mayberry, 2008; Pohlen et al., 2022). This problem is exacerbated in Mars' frequent dust storms. The accumulation and erosion of dust on these surfaces can gradually degrade the functionality of solar panels, thermal control systems, optical equipment, spacesuits, and machinery, and in some cases, may even jeopardize astronaut safety. By employing EICPP to stabilize the regolith, future space missions can enhance both safety and efficiency. Unlike biocements that rely on live microorganisms, EICPP uses crude enzymes, significantly reducing the risk of microbial contamination in the confined ecosystems of extraterrestrial environments, thus offering significant biosecurity advantages. Crude enzymes can withstand large temperature changes (Miao et al., 2020a, 2020b; Omarov et al., 2023; Wang et al., 2023a, 2023b), which makes them more advantageous than biocement that relies on microbial culture in dealing with extreme environments.

Moreover, the plants used to extract crude enzymes, such as soybeans, could bring additional benefits to interstellar bases. In space, oxygen is a precious resource that must be meticulously managed and prioritized for life support systems. During plant growth, they can provide valuable oxygen for life support. In addition, compared to microbe-based biocement, enzyme-driven biogeochemical processes do not require oxygen (Jiang et al., 2016; Li et al., 2018; Xiao et al., 2022), making them ideal for applications on the Moon and Mars. Furthermore, plants can serve as a food source, enhancing the sustainability of extraterrestrial bases. Even after enzyme extraction, the protein-rich soybean residue can be repurposed as a nutritional supplement (Guan et al., 2023), further closing the loop on resource utilization. Beyond their functional contributions, plants can also provide psychological and emotional support. Research has shown that in closed environments, the presence of plants helps alleviate astronaut stress, mitigates the psychological challenges of isolation and harsh conditions in long-duration missions, and improves overall mental health, contributing to team morale (De Micco, 2023; De Pascale, 2021; Odeh & Guy, 2017; Smith, 2024; Zhao et al., 2023). However, further research is necessary to better understand the physical and chemical properties of lunar and Martian regolith (Colwell et al., 2007; Ding et al., 2024; Engelschiøn et al., 2020; Papike et al., 1982; Zakharov et al., 2020), and how these characteristics might influence the EICPP process. Consolidation

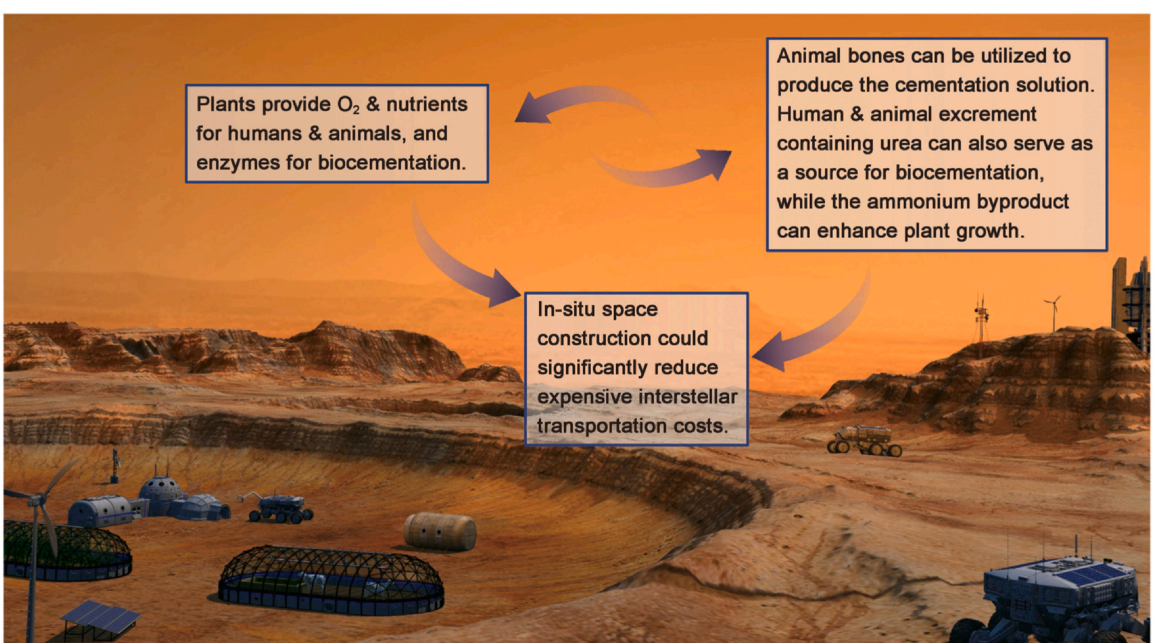

Fig. 4. Vision of using EICPP biocement for extraterrestrial construction.

experiments should also be conducted under microgravity conditions, as regolith composition and behavior could impact the efficiency of biochemical processes. Understanding these variables is crucial for optimizing this technology. Additionally, larger-scale experiments simulating lunar and Martian conditions, alongside life cycle assessment (LCA) analyses, are needed to validate the feasibility and cost-effectiveness of EICP for actual space missions, paving the way for its implementation in future extraterrestrial colonization efforts.

4. Conclusion

This study demonstrates the feasibility of kitchen waste bone-driven crude enzyme-induced calcium phosphate precipitation under both normal gravity and microgravity conditions, offering promising insights for space biocementation applications. The experimental results showed no significant differences in the degree of precipitation, reaction uniformity, or microstructural characteristics between normal and microgravity environments. This suggests that EICPP is a viable solution for extraterrestrial construction, particularly during the expansion phase of human settlement in space. The utilization of waste materials, such as animal bones, for the extraction of calcium and phosphorus further highlights the potential for this technology to support sustainable resource management in extraterrestrial environments. By integrating local resources and waste materials, EICPP can reduce the need for material transport from Earth, lowering costs and minimizing environmental impacts associated with frequent space missions. Moreover, the use of a neutral pH system reduces the risk of ammonia emissions, enhancing the biosecurity of enclosed space habitats. This study underscores the potential of EICPP not only for space base construction but also for addressing waste disposal challenges, contributing to the overall sustainability and self-sufficiency of future extraterrestrial colonies.

CRediT authorship contribution statement

Zhen Yan: Writing – review & editing, Writing – original draft, Visualization, Investigation, Funding acquisition. **Kazunori Nakashima:** Resources. **Chikara Takano:** Resources. **Satoru Kawasaki:** Resources, Funding acquisition, Data curation.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data statement

All the experimental data that support the findings of this study are available from the corresponding author upon reasonable request through email.

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